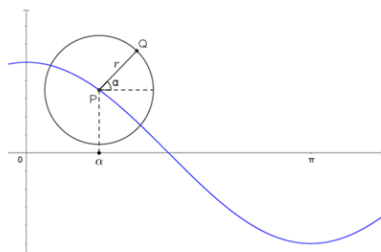


Question 24 - Stage 2



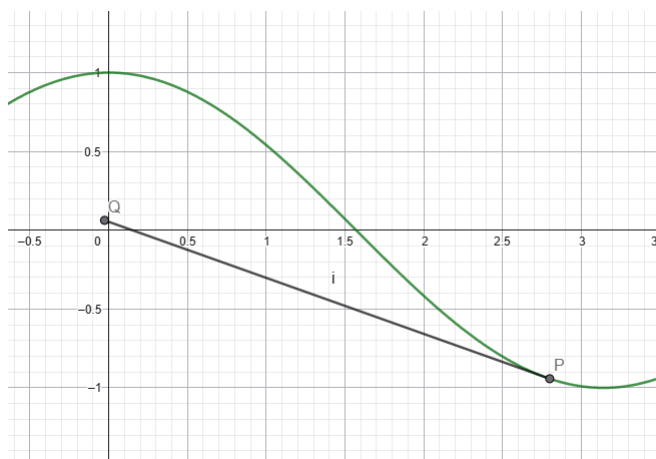
A point P moves along the curve $y = \cos x$. P is the centre of a circle of radius r , ($r > 0$).

When P has coordinates $(\alpha, \cos \alpha)$, the vector $\vec{PQ} = \begin{pmatrix} r \cos \alpha \\ r \sin \alpha \end{pmatrix}$ determines the position of point Q on the circle.

Given that Q passes through $(0, 0)$ for a value of α where $0 < \alpha < 2\pi$, find the value of r (to 3d.p.)

Working

$$\begin{pmatrix} \alpha \\ \cos(\alpha) \end{pmatrix} + \begin{pmatrix} r \cos(\alpha) \\ r \sin(\alpha) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$



α needs to be facing towards the origin meaning, by inspection, α needs to be top left hand quadrant of a circle when P is below and to the right of the origin, this means...

$$\begin{aligned} \frac{\pi}{2} < \alpha < \pi \\ -1 < \cos(\alpha) < 0 \\ 0 < \sin(\alpha) < 1 \end{aligned}$$

$$r = -\frac{\alpha}{\cos(\alpha)}$$

$$r = -\frac{\cos(\alpha)}{\sin(\alpha)}$$

$$\frac{\alpha}{\cos(\alpha)} = \frac{\cos(\alpha)}{\sin(\alpha)}$$

$$\alpha \sin(\alpha) = 1 - \sin^2(\alpha)$$

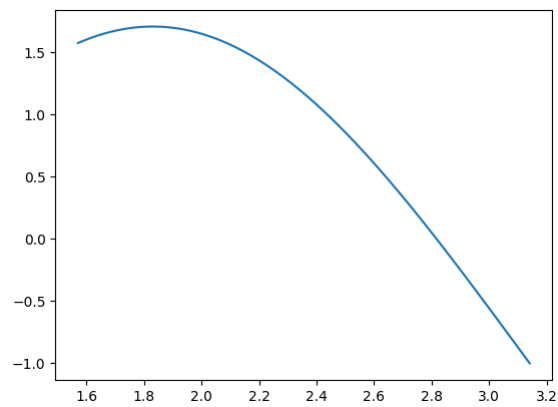
$$\sin^2(\alpha) + \alpha \sin(\alpha) - 1 = 0$$

```
In [54]: from math import pi, sin
import matplotlib.pyplot as plt
import numpy as np
from pyTools import sign

alphaMin = pi/2
alphaMax = pi

xCoords = np.linspace(alphaMin, alphaMax, 1000)
fx = lambda alpha: sin(alpha)**2 + alpha*sin(alpha) - 1
yCoords = [fx(x) for x in xCoords]

plt.plot(xCoords, yCoords)
plt.show()
```



```
In [55]: # Starting from to the right near the correct value to ensure it doesn't get stuck near a turning point
alpha = 2.8
lastDistance = 999
precision = 0.00001
precise = False

while not precise:
    output = fx(alpha)
    distance = abs(output)
    if distance <= precision:
        precise = True
    else:
        alpha += precision * sign(lastDistance-distance)
        lastDistance = distance

print(f"Approximate alpha: {alpha}")
```

Approximate alpha: 2.8170400000001115

$$r = -\frac{\alpha}{\cos(\alpha)}$$

$$r \approx 2.972$$